Riad Mahmud Chowdhury

riadc916@gmail.com | riadmahmud@stmail.ntu.edu.cn
riadmahmud.github.io | Google Scholar | ORCID: 0009-0000-8946-4458

CAREER OBJECTIVE

My goal is to expand my research and technical capabilities in Data Science, Machine Learning (ML), and Cybersecurity across the end-to-end data lifecycle—from data preprocessing and exploratory analysis to online learning and model deployment. Building upon my prior research in game-theoretic moving-target defense and cyber-physical network security, I aim to develop robust, scalable defense algorithms while investigating methods to make machine learning workflows more energy-efficient and environmentally sustainable.

RESEARCH INTERESTS

Machine Learning (ML), **Data Science**, **Cybersecurity** (Cyber-Physical Systems Security, Moving-Target Defense, Online Convex Optimization & Game Theory, Energy-Efficient ML)

EDUCATION

Nantong University, School of Artificial Intelligence and Computer Science — **Nantong, Jiangsu, China**
Bachelor of Science Degree in Computer Science and Technology — 2023.09.10 – Present (Exp. 2027)
CGPA: **4.00/4.00** (Completed 6th Semester / 144 Credits)

Chattogram Bandar College — **Chattogram, Bangladesh**
Higher Secondary Certificate (HSC) in Science — 2018 – 2020
Result: GPA: **4.87 / 5.00**

Mandarin Language Proficiency (HSK Level 4) — **Nantong University, China**
Result: **220 / 300** — 2025.12

SKILLS SUMMARY

- **Programming:** Python, R, JavaScript, C#, Java, SQL, HTML/CSS
- **Data Science & AI/ML:** Machine Learning, Online Convex Optimization (Entropic Mirror Descent), Parametric Linear Programming (LP), Game-Theoretic Modeling, Statistical Hypothesis Testing (χ^2 Detector), Exploratory Data Analysis (EDA), Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Tidyverse, Ggplot2
- **Tools & Platforms:** pandapower, SciPy / LP Solvers, Visual Studio Code, Google Colab, Jupyter Notebook, Git, GitHub, LaTeX, RStudio, Zotero, Microsoft Office
- **Languages:** Bengali (Native), English (Advanced / Fluent), Mandarin (Intermediate)

CONFERENCE PUBLICATIONS

- **International Conferences:**
  - **[IEEE CSDE'26]:** Md Mostofa Nurannabi Shakil, **Riad Mahmud Chowdhury**, Tanvir Hossain, Md. Alamgir Hossain, Md. Samiul Islam, and Md. Shafiqul Islam, "Provable Moving-Target Defense for Cyber-Physical Networks: From the Security–Performance Frontier to No-Regret Defense," in Proceedings of the 12th IEEE Asia-Pacific Conference on Computer Science and Data Engineering (IEEE CSDE 2026), Cox's Bazar, Bangladesh, November 1–3, 2026. [Accepted].

RESEARCH EXPERIENCE

Research Contributor & Co-Author (Cyber-Physical Systems & Data Analysis) — IEEE CSDE 2026

- **Experimental Modeling & Simulation:** Instantiated and validated a game-theoretic Moving-Target Defense (MTD) framework on **IEEE 14-bus (34 measurements)** and **IEEE 118-bus (304 measurements)** power grids using pandapower across **12 D-FACTS reactance configurations** (±60% range).
- **Security & Optimization Analysis:** Evaluated False-Data-Injection (FDI) stealth against DC Weighted-Least-Squares (WLS) state estimation and χ^2 residual detectors (5% false-alarm rate), empirically estimating confusion kernels (K) and validating parametric Linear Programming (LP) security–cost frontiers.
- **Algorithmic Evaluation & Scalability Impact:** Benchmarked entropic online mirror descent achieving sublinear $O(\sqrt{T \log N})$ regret over 6,000 epochs and demonstrated favorable grid scalability by reducing off-diagonal attack stealth from **0.14 to 0.04** and lowering the matrix-game security floor val(K) from **0.21 to 0.13**.

PROJECTS & SOFTWARE DEVELOPMENT

• **[Banking-Web-C-] Online Banking Web System** Dec 2025

Architected a secure full-stack web application using C# implementing robust object-oriented transaction logic, customer account management, and relational database connectivity. Enforced backend data integrity, input validation, and modular architecture for reliable financial operations.

• **[lego-data-analysis] Multi-Decade Structural Dataset Insights** 2026

Analyzed **15,710 LEGO sets** (1949–2021) and **596 themes** in Python (Pandas, Matplotlib) by merging relational schemas. Identified **135 unique colors** (107 opaque, 28 transparent), quantified an **18× surge in set complexity** from **12.36 average parts** (1954) to **221.84 parts** (2017; peaking at 9,987 parts), and established Star Wars as the top franchise with **753 sets**.

• **[StackOverflow-Language-Trends] Developer Adoption Analytics** 2026

Built a time-series pipeline in Python (Pandas, Matplotlib) using SQL-extracted StackExchange data (**1,991 monthly records** across **14 programming languages**, 2008–2020). Applied pivot tables and 12-month rolling-mean smoothing, revealing **JavaScript** as the cumulative all-time leader (**2,056,510 posts**) while demonstrating **Python's** rapid trajectory to **#1 monthly active language** (~34,478 posts/month by 2020).

• **[college-salaries-analysis-pandas] Post-Graduation Compensation Modeling** 2026

Evaluated PayScale compensation data across **50 undergraduate majors** categorized into STEM, Business, and HASS using Pandas. Identified Physician Assistant as the top starting major (**$74,300**), Chemical Engineering as the highest mid-career median earner (**$107,000**), Nursing as the lowest-risk career (**$50,700** 10th–90th percentile spread) vs. Economics with the highest upside (**$210,000** 90th percentile), and proved **STEM majors** lead in group-level mid-career median pay (**$90,812.50**).

• **[Superstore-Sales-Profitability-Analysis] Enterprise Retail Analytics** 2026

Engineered a modular R analytics workflow (01_clean_the_dataset.R, 02_superstore_analysis.R using Tidyverse & Ggplot2) to transform raw retail records into cleaned datasets and automated visual/tabular reports. Quantified sub-category profit/loss margins, regional shipping mode costs, order priority revenue distributions, and customer segment discount sensitivity, uncovering how aggressive discounting erodes net profitability.

CERTIFICATIONS & PROFESSIONAL TRAINING

• **Data, Data, Everywhere — Google Data Analytics Professional Certificate** Coursera | 2026

Credential: Certificate Verification Link [Verified / Completed]

• **100 Days of Code: The Complete Python Pro Bootcamp** Udemy | 2026

Credential: Certificate Verification Link [Verified / Completed]

ACADEMIC HONORS & EXTRA-CURRICULARS

- **Perfect Academic Standing (CGPA 4.00 / 4.00):** Ranked in top tier throughout all completed semesters at Nantong University, School of Artificial Intelligence and Computer Science (2023 – Present).
- **Academic Mentorship:** Assisted peer student groups in scientific Python environments, structured notebook practices, and data visualization setups.

ACADEMIC REFERENCES

Xiaolin Ju
Associate Professor
School of Artificial Intelligence and Computer Science
Nantong University, Nantong, Jiangsu, China
Email: ju.xl@ntu.edu.cn

Wan Jie
Associate Professor, Master's Supervisor
Department of Artificial Intelligence
School of Artificial Intelligence and Computer Science
Nantong University, Jiangsu, China
Email: jiewan@ntu.edu.cn

Haochen Wang (王皓晨)
Deputy Director of the Department of Software Engineering
School of Artificial Intelligence and Computer Science
Nantong University, Nantong, Jiangsu, China
Email: haochenwang@ntu.edu.cn